

Switch (n) {

Case 1: cout << "Monday" << endl;
break;

Case 2: cout << "Tuesday" << endl;
break;

Case 3: cout << "Wednesday" << endl;
break;

Case 4: cout << "Thursday" << endl;
break;

Case 5: cout << "Friday" << endl;
break;

Case 6: cout << "Saturday" << endl;
break;

Case 7: cout << "Sunday" << endl;
break;

default: cout << "Invalid Input" << endl;

}

return 0;

}

• Number System Conversion:-

(A) Decimal To Any Number System

(i) Keep dividing the number by the base of required number system and store the remainders.

(ii) The converted number is obtained by reading remainders from bottom to top.

ex → Convert $(13)_{10}$ to Binary

2	13	
2	6	→ 1
2	3	0
2	1	1
	0	1

Remainders

So,

$$(13)_{10} = (1101)_2$$

we keep dividing till the number becomes zero.



(B) Any Number System To Decimal

(i) Multiply each bit by the corresponding power of base of current number system.

(ii) Power starts from 0 on the rightmost side.

ex $\rightarrow$ Convert $(1101)_2$ to decimal

$$(1101)_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 8 + 4 + 0 + 1$$

$$= (13)_{10}$$

Q) WAP to convert integer to binary

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter the number: ";
```

```
    cin >> n;
```

```
    int rem, ans=0, multi=1
```

$\rightarrow$ Multi 1 se shuru hoga

```
    while(n!=0) {
```

```
        rem = n%2;
```

```
        n /= 2;
```

```
        ans = ans * 10 + rem;
```

```
        multi *= 10;
```

```
    }
    cout << "Binary Form: " << ans << endl;
```

```
    return 0;
```

```
}
```

Q) WAP to Convert Binary To Decimal

```
#include <iostream>
using namespace std;
```

```
int main () {
```

```
    int num;
```

```
    cout << "Enter the binary number: ";
```

```
    cin >> num;
```

int rem, ans=0, multi=1; → Multi 1 se Shuru hoga kyu ki $2^0 = 1$ hota hai.

```
    while (num != 0) {
```

```
        rem = num % 10;
```

```
        num /= 10;
```

```
        ans += rem * multi;
```

```
        multi *= 2;
```

```
    }
    cout << "Decimal Form: " << ans << endl;
```

```
    return 0;
```

• IMPORTANT → When converting to decimal, multi is multiplied by existing base (Pehle wale Number System ka base)

When converting from decimal, multi is multiplied by powers of 10
multi *= 10;

Q) WAP to Convert decimal to Octal

```
#include <iostream>
using namespace std;
```

```
int main () {
```

```
    int num;
```

```
    cout << "Enter the decimal number: ";
```

```
    cin >> num;
```

```
    int rem, ans=0, multi=1;
```

```
while (num != 0) {
```

```
    rem = num % 8;
```

```
    num /= 8;
```

```
    ans += rem * multi;
```

```
    multi *= 10;
```

```
}
```

```
cout << " Octal Form: " << ans << endl;
```

```
return 0;
```

```
}
```

Q) WAP to Convert Binary Form to Octal Form:

```
#include <iostream>
```

```
using namespace std;
```

```
int main () {
```

```
    int num;
```

```
    cout << "Enter the number: ";
```

```
    cin >> num;
```

// Converting to decimal

```
int rem, ans=0, multi=1;
```

```
while (num != 0) {
```

```
    rem = num % 10;
```

```
    num /= 10;
```

```
    ans += rem * multi;
```

```
    multi *= 2;
```

```
}
```

// Decimal to Octal

```
int rem2, ans2=0, multi2=1;
```

```
while (ans2 != 0) {
```

```
    rem2 = ans2 % 8;
```

```
    ans2 /= 8;
```

```
    ans2 += rem2 * multi2;
```

```
    multi2 *= 10;
```

```
}
```

```

cout << "Octal Form: " << ans2 << endl;
return 0;

```

Q) WAP to Convert decimal numbers to binary numbers using for loop.

```

#include <iostream>
using namespace std;

```

```

int main() {

```

```

    int num;

```

```

    cout << "Enter the decimal number: ";

```

```

    cin >> num;

```

```

    int rem, ans = 0, multi = 1;

```

```

    for (; num != 0; ) {

```

```

        rem = num % 2;

```

```

        num /= 2;

```

```

        ans = ans * multi + rem;

```

```

        multi *= 10;
    }

```

Initialization Part is empty
 condition → num != 0
 Update part is empty.
 So, it is equivalent to
 while (n != 0)

```

    cout << "Binary Equivalent: " << ans << endl;
    return 0;
}

```

Q) WAP to Convert Octal to Binary

```

#include <iostream>
using namespace std;

```

```

int main() {

```

```

    int num;

```

```

    cout << "Enter the octal number: ";

```

```

    cin >> num;

```

// Octal To Decimal

```

    int rem, ans = 0, multi = 1;

```

```
while (num != 0) {
```

```
    rem = num % 10;
```

```
    num /= 10;
```

```
    ans = ans * multi;
```

```
    multi *= 8;
```

```
}
```

```
int num2 = ans;
```

```
// Decimal to Binary Conversion
```

```
int rem2, ans2 = 0, multi2 = 1;
```

```
while (num2 != 0) {
```

```
    rem2 = num2 % 2;
```

```
    num2 /= 2;
```

```
    ans2 = ans2 * multi2;
```

```
    multi2 *= 10;
```

```
}
```

```
cout << "Binary Equivalent: " << ans2 << endl;
```

```
return 0;
```

```
}
```

Q) Given an integer num, repeatedly add all its digits until the result has only one digit, and return it.

```
#include <iostream>
```

```
using namespace std;
```

```
int main () {
```

```
    int num;
```

```
    cout << "Enter the number: ";
```

```
    cin >> num;
```

```
    int sum = 0, id;
```

```
    while (num > 9) {
```

```
        sum = 0
```

Sum must be reset to 0 for every digit-sum iteration, otherwise previous sum accumulate and give wrong results.